

Relational
Programming

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Given $S \downarrow i = i^{th}$ word.

$S =$

1	'to'
2	'be'
3	'or'
4	'not'
5	'to'
6	'be'

Want $F \downarrow w = \text{freq. of word } w$.

$F =$

'to'	2
'be'	2
'or'	1
'not'	1

First reverse columns

$$S^{-1} =$$

‘to’	1
‘be’	2
‘or’	3
‘not’	4
‘to’	5
‘be’	6

Form the “unit image” of the table

$$\text{unimage } T^{-1} =$$

‘to’	{1, 5}
‘be’	{2, 6}
‘or’	{3}
‘not’	{4}

Send this through the size function

'to'	{1, 5}
'be'	{2, 6}
'or'	{3}
'not'	{4}

|

{}	0
{1}	1
{4}	1
{2, 6}	2
{4, 8}	2
{4, 6, 7}	3
.	.
:	:

||

'to'	2
'be'	2
'or'	1
'not'	1

The resulting program is:

$$F \equiv \text{unimage } S^{-1} \mid \text{size}$$

We can turn it into a function definition to compute the frequency table for any string by:

$$\text{freq } S \equiv \text{unimage } S^{-1} \mid \text{size}$$

RPL INTERPRETER ON LINE!!

```
?> S == (list "to" "be" "or" "not" "to" "be")
```

```
?> (S sup -1]
```

```
( rel ( be 6 ) ( to 5 ) ( not 4 )  
  ( or 3 ) ( be 2 ) ( to 1 ) )
```

```
?> (unimage (S sup -1]
```

```
( set ( be ( set 6 2 ) ) ( to ( set 5 1 ) )  
  ( not ( set 4 ) ) ( or ( set 3 ) )  
  ( be ( set 6 2 ) ) ( to ( set 5 1 ) ) )
```

```
?> ((unimage (S sup -1))  
rp size]
```

```
( rel ( be 2 ) ( to 2 ) ( not 1 ) ( or 1 )  
  ( be 2 ) ( to 2 ) )
```

```
?> freq S == ((unimage (S sup -1)) rp size]
```

```
?> (freq (list "to" "be" "or" "not" "to" "be"]
```

```
( rel ( be 2 ) ( to 2 ) ( not 1 ) ( or 1 )  
  ( be 2 ) ( to 2 ) )
```

```
?> done
```

Pretty Printed Form of Program

```
?> freq S ≡ unimage S-1 | size
```

```
?> freq <"to", "be", "or", "not", "to", "be">
```

```
{"be" : 2, "to" : 2, "not" : 1, "or" : 1}
```

```
?> done
```

RPL INTERPRETER ON LINE!!

```
?> F == (rel (124 :  
  (rel ("N" : "John") ("R" : 10) ("H" : 100)))  
(118 :  
  (rel ("N" : "Bill") ("R" : 15) ("H" : 120)))  
(207 :  
  (rel ("N" : "Sally") ("R" : 14) ("H" : 115]
```

```
?> U == (rel (118 : 6) (124 : 40) (207 : 40]
```

```
?> (F # U]
```

```
( rel ( 124 ( rel  
  ( 1 ( rel ( N John ) ( R 10 ) ( H 100 ) ) )  
  ( 2 40 ) ) ) )  
( 118 ( rel  
  ( 1 ( rel ( N Bill ) ( R 15 ) ( H 120 ) ) )  
  ( 2 6 ) ) ) )  
( 207 ( rel  
  ( 1 ( rel ( N Sally ) ( R 14 ) ( H 115 ) ) )  
  ( 2 40 ) ) ) ) )
```

```
?> sumhrs == ((op +) o ((rsec sel "H") || I]
```

```
?> upd == ((F # U) rp  
  (as o ((lsec "H" ,) o sumhrs]
```

```
?> F' == ((upd # F) rp (op ;]
```

```
?> val F'
```

```
( rel ( 124  
  ( rel ( H 140 ) ( N John ) ( R 10 ) ) )  
( 118  
  ( rel ( H 126 ) ( N Bill ) ( R 15 ) ) )  
( 207  
  ( rel ( H 155 ) ( N Sally ) ( R 14 ) ) ) )
```

```
?> done
```

```
(F == (rel (124 :  
  (rel ("N" : "John") ("R" : 10) ("H" : 100)))  
  (118 :  
    (rel ("N" : "Bill") ("R" : 15) ("H" : 120)))  
  (207 :  
    (rel ("N" : "Sally") ("R" : 14) ("H" : 115))))))
```

```
(U == (rel (118 : 6) (124 : 40) (207 : 40)))
```

```
(sumhrs == ((op +) o ((rsec sel "H") || I)))
```

```
(upd == ((F # U) rp  
  (as o ((lsec "H" ,) o sumhrs))))
```

```
(F' == ((upd # F) rp (op ;)))
```

$$F \equiv \{124 : \{"N" : "John", "R" : 10, "H" : 100\}, \\ 118 : \{"N" : "Bill", "R" : 15, "H" : 120\}, \\ 207 : \{"N" : "Sally", "R" : 14, "H" : 115\}\}$$

$$U \equiv \{118 : 6, 124 : 40, 207 : 40\}$$

$$\text{sumhrs} \equiv [+] \circ ([\downarrow H] \parallel \mathbf{I})$$

$$\text{upd} \equiv (F \# U) \mid (\text{as} \circ ([H,] \circ \text{sumhrs}))$$

$$F' \equiv (\text{upd} \# F) \mid [;]$$

-- DFA Minimization

-- Utility Functions

$$1st \equiv [\downarrow 1]$$

$$2nd \equiv [\downarrow 2]$$

$$r \perp s \equiv r \mid s \mid r^{-1}$$

$$\sigma f \equiv (f \circ (1st \bar{,} (\mathcal{E} \circ 2nd))) \bar{,} \\ ([\backslash] \circ (\mathbf{I} \bar{,} (\text{un} \circ \mathcal{E}) \circ 2nd))$$

$$f \rho i \equiv 1st \circ (\sigma f \text{ while } ([\neq \emptyset] \circ 2nd)) \circ [i ,]$$

$$\cup \equiv [\cup] \rho \emptyset$$

-- Example DFA

$$\Sigma \equiv \{1, 2\}$$

$$T \equiv \{1 : \{10 : 10, 20 : 20\}, 2 : \{10 : 30, 20 : 30\}\}$$

$$Q \equiv \{10, 20, 30\}$$

$$F \equiv \{30\}$$

-- Minimization

$$Q^2 \equiv Q \times Q$$

$$n \equiv \text{size } Q^2$$

$$R_0 \equiv F \times (Q \setminus F)$$

$$\psi R \equiv R \cup \bigcup(\text{rng } (T \mid [\perp R]))$$

$$R_\infty \equiv \psi^n R_0$$

$$R_- \equiv (Q^2 \setminus R_\infty) \cap (Q^2 \setminus R_\infty^{-1})$$

$$\text{eclass} \equiv [R_- \text{ unimg}]$$

$$\text{equiv} \equiv [\text{eclass img}]$$

$$Q_- \equiv \text{equiv } Q$$

$$T_- \equiv T \mid [\text{eclass } \$]$$

$$F_- \equiv \text{equiv } F$$

-- Gaussian Elimination

-- Utility Functions

$\text{con } k \equiv \lambda x. k$

$\text{transmap } f \equiv [\mid f] \circ [\#]$

$\text{vecdif} \equiv \text{transmap } [-]$

$\text{scaprod } \langle k, v \rangle \equiv v \mid [k \times]$

$\text{outerprod } \langle u, v \rangle \equiv u \mid (\text{scaprod} \circ [, v])$

$\text{matdif} \equiv \text{transmap } \text{vecdif}$

$\text{column } \langle M, k \rangle \equiv M \mid [\downarrow k]$

$\text{unit } \langle M, k \rangle \equiv \langle 1, \dots, \text{size } M \rangle \mid$
 $[[= k] \rightarrow \text{con } 1; \text{con } 0]$

$\text{diag } \langle M, k \rangle \equiv M \downarrow k \downarrow k$

$f \text{ for } S \equiv [@ S] \circ [f \$]$

-- Gaussian Elimination

$$V \equiv \text{scaprod} \circ (([1.0 \ /] \circ \text{diag}) \circ \text{vecdif} \circ (\text{column} \circ \text{unit}))$$
$$\text{elim} \equiv \text{matdif} \circ ([\downarrow \ 1] \circ (\text{outerprod} \circ (V \circ [\downarrow])))$$
$$\text{Gauss } M \equiv (\text{elim for } < 1, \dots, \text{size } M >) M$$

-- Example Matrix

$$M \equiv \begin{pmatrix} 3 & 9 & 33 \\ 2 & -1 & 1 \end{pmatrix}$$